

CODE	COURSE NAME	INFORMATION	IT	CH	NO	CREDITS
221EIT003	ETHICAL HACKING	PROGRAM ELECTIVE I	3	0	0	3

Preamble: The students are expected to understand the basic concepts of Ethical Hacking and Penetration testing.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Perform system security vulnerability testing
CO 2	Learn various hacking methods
CO 3	Perform system vulnerability exploit attacks
CO 4	Produce a security assessment report
CO 5	Recognize tools and methods to carry out penetration testing.

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1:An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2:An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3:An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor's program

PO4:An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards

PO5:An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tools to model, analyze and solve practical engineering problems.

PO6:An ability to engage in lifelong learning for the design and development related to the stream-related problems taking into consideration sustainability, societal, ethical, and environmental aspects

PO7:An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4 INFC	PO 5 INFORMATION	PO 6 TECHNOLOGY	PO 7-IT1
CO 1	✓		✓	✓	✓	✓	
CO 2	✓		✓	✓	✓	✓	
CO 3	✓		✓	✓	✓	✓	
CO 4	✓	✓	✓	✓	✓	✓	
CO 5	✓	✓	✓	✓	✓	✓	

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	80
Analyse	20
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

The evaluation shall only be by employing application-based, analysis-based, or design-based questions (for both internal and end-semester examinations)

Continuous Internal Evaluation Pattern:

The evaluation shall only be based on application, analysis or design-based questions (for both internal and end-semester examinations).

Continuous Internal Evaluation: 40 marks

- Preparing a review article based on peer-reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- Course based task / Seminar/ Data collection and interpretation : 15 marks
- Test paper (1 number) : 10 marks

Test paper shall include a minimum of 80% of the syllabus.

Course-based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.

There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum of one question from each module of which student should answer any five. Each question can carry 7 marks

The total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example, if the average end-semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.



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M.TECH DEGREE EXAMINATION
.....SEMESTER
Branch:

Time: 2½ Hours, Maximum marks: 60

PART A (Answer all questions. Each question carries 5 marks)

1. Explain the terms Threat, Vulnerability, and Risk.
2. What are the different methods for managing insider threats?
3. What do you mean by 'Vulnerability Scanner'? Draw the scanner and briefly explain
4. Explain in detail computer fraud classification according to the data processing model?
5. Is there any method to demonstrate whether network infrastructure is adequate or not? Justify your answer.

PART B (Answer any five questions. Each question carries 7 marks)

6. Being an expert in hacking, you are asked to analyze the free, open-source software being used to run FTP services on a server. You noticed that there is an excessive number of `gets()` and `get()` on the source code. These C++ functions do not check bounds. What kind of attack is this program susceptible to? Comment on your recommendation to prevent this attack.
7. Explain the type of attack faced by the stack data structure. Also, explain the uses of two pointers used in a stack.
8. Illustrate how SQL injection attacks are done by attackers.
9. Discuss some architecture strategies for fraud prevention with respect to computer network-enabled system
10. Explain IT infrastructure.
11. An attacker uses a distributed group of computers to shut down a single machine or network making it inaccessible to its intended users? Explain the attack with the help of a neat diagram. What are the ways to prevent this attack?
12. Illustrate the process involved in understanding the inherent risk profile of a business with relevant tables and assumptions

Syllabus INFORMATION TECHNOLOGY-IT1

Module I: (8 Hours)

Introduction: CIA Triads, Hacking Impact, The Hacker Framework: Planning the test, Sound operations, Reconnaissance, Enumeration, Vulnerability Analysis, Exploitation, Final Analysis, and Deliverable. Integration Information Security Models: Computer Security, Network Security, Service Security, Application Security. Security Architecture Information Security Program: The Process of Information Security, Component Parts of Information Security Program, Risk Analysis, and Ethical Hacking.

Module II: (8 Hours)

The Business Perspective: Business Objectives, Security Policy, Previous Test Result. Business Challenges Planning for a Controlled Attack: Inherent Limitations Imposed Limitations, Timing Everything, Attack Type, Source Point, Required Knowledge, Multi-Phased Attack, Teaming and Attack Structure, Engagement Planner, The Right Security Consultant, The Tester, Logistics, Intermediates, Law Enforcement.

Module III: (8 Hours)

Preparing for a Hack: Technical Preparation, Managing the Engagement, Reconnaissance: Social Engineering, Physical Security, and Internet Reconnaissance.

Module IV: (8 Hours)

Enumeration: Enumeration Techniques, Soft Objective, Looking Around or Attack, Elements of Enumeration. Preparing for the Next Phase Exploitation: Intuitive Testing, Evasion, Threads and Groups, Operating Systems, Password Crackers, Rootkits, Applications, Wardialing, Network, Services, and Areas of Concern

Module V: (8 Hours)

Deliverable: The Deliverable, The Document, Overall Structure, Aligning Findings. Presentation Integration: Integrating the Results, Integration Summary, Mitigation, Defense Planning, Incident Management, Security Policy

Reference Books

1. Hacking: The Art of Exploitation Jon Erikson 2nd edition
2. The Basics of Hacking and Penetration Testing Patrick Engebretson 2nd edition
3. The Hacker Playbook 2 Peter Kim 1st edition
4. Penetration Testing Georgia Weldman 1st edition.

5. The Web Application Hacker's Handbook Dafydd Stuttard 2nd edition.
6. Stuart McClure, Joel Scambray and George Invernizzi, "Information Technology Security Secrets & Solutions", Tata McGraw-Hill Publishers, 2010 (6th Edition)
7. Ben Smith and Brian Komer, "Microsoft Windows Security Resource Kit", Prentice Hall of India, 2010

Syllabus and Course Plan (For 3 credit courses, the content can be for 40 hrs, and for 2 credit courses, the content can be for 26 hrs. The audit course in the third semester can have content for 30 hours).

No	Topic	No. of Lectures (40 Hours)
1	Module 1	8 Hours
1.1	Introduction: CIA Triads, Hacking Impact.	2
1.2	The Hacker Framework: Planning the test, Sound Operations, Reconnaissance, Enumeration, Vulnerability Analysis, Exploitation, Final Analysis, and Deliverable.	2
1.3	Integration Information Security Models: Computer Security, Network Security, Service Security, Application Security	2
1.4	Security Architecture Information Security Program: The Process of Information Security, Component Parts of Information Security Program, Risk Analysis, and Ethical Hacking	2
2	Module 2	9 Hours
2.1	The Business Perspective: Business Objectives, Security Policy, Previous Test Result	2
2.2	Business Challenges Planning for a Controlled Attack: Inherent Limitations Imposed Limitations,	2
2.3	Timing Everything, Attack Type, Source Point, Required Knowledge, Multi-Phased Attack	1
2.4	Teaming and Attack Structure, Engagement Planner,	2
2.5	The Right Security Consultant, The Tester, Logistics, Intermediates, Law Enforcement.	2
3	Module 3	7 Hours
3.1	Preparing for a Hack: Technical Preparation,	1
3.2	Managing the Engagement	2
3.3	Reconnaissance: Social Engineering	2
3.4	Physical Security, and Internet Reconnaissance.	2
4	Module 4	8 Hours
4.1	Enumeration: Enumeration Techniques, Soft Objective,	1

4.2	LookingAround or Attack, Elements of Enumeration.	2
4.3	Preparing for the Next Phase Exploitation: Intuitive Testing, Evasion, Threads and Groups, Operating Systems, INFORMATION TECHNOLOGY-IT	2
4.4	Password Crackers, Rootkits,Applications	2
4.5	Wardialing, Network, Services and Areas of Concern.	1
5	Module 5	8 Hours
5.1	Deliverable: The Deliverable, The Document, Overall Structure, Aligning Findings	2
5.2	Presentation Integration: Integrating the Results, Integration Summary	2
5.3	Mitigation, Defense Planning	2
5.4	Incident Management, Security Policy	2

